

NERMINE MELKI AMDOUNI

Data Scientist | Machine Learning & AI

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SUMMARY

Data Scientist and Machine Learning Engineer building production-ready, LLM-powered AI applications, deep learning models, and data pipelines with Python and SQL.

EXPERIENCE

Data Consulting Intern, EY AMC (Ernst & Young) February 2025 – July 2025

- Engineered a real-time speech translation system (English, French, Spanish), benchmarking over 20 models across speech-to-text, translation, and text-to-speech.
- Designed a multi-threaded Flask/WebRTC pipeline (speech recognition, translation, GPT-4 correction, text-to-speech), cutting latency to 3.5s using Faster Whisper and MarianMT.
- Implemented GPT-4 text cleaning, live transcription UI, and PDF/MP3 export, validating performance under real-time load with pytest.

Data Analyst / Development Intern, NEXT STEP June 2024 – August 2024

- Developed a full-stack monitoring web app using React, Node.js, and Flask to track cloud platform performance in real time via SQL and MongoDB queries.
- Built automated data collection and structuring pipelines centralized in a MongoDB database, managed via Git.
- Collaborated with cross-functional teams to analyze cloud performance metrics and streamline real-time data tracking workflows.

Developer / Data Scientist (Academic Collab.), Swiss Premium Negoce January 2024 – May 2024

- Developed a full-stack app using Angular, Flask, and MongoDB integrating a Mistral 7B LLM to automate data extraction from 1,577 records and 25+ multilingual documents.
- Designed data cleaning and annotation pipelines used to train and refine the extraction model.
- Built interactive Power BI dashboards to track operational performance and key business KPIs.

PROJECTS

Network Intrusion & Anomaly Detection – NSL-KDD Dataset (Team Project, 4 members, CRISP-DM methodology) January 10 – 31, 2024

- Ran a two-part ML project (CRISP-DM) on NSL-KDD: supervised classification and unsupervised anomaly detection.
- Benchmarked 6 classifiers (Decision Tree, Random Forest, KNN, SVM, Logistic Regression, AdaBoost) with GridSearchCV, reaching 86.4% test accuracy.
- Compared 4 clustering algorithms (KMeans, DBSCAN, GMM, Hierarchical), selecting GMM as best model via Silhouette Score.
- Packaged the model into a Flask app with pickle serialization, visualized using Matplotlib/Seaborn.

PMBOK Knowledge Graph & RAG Chatbot September 20 – December 15, 2024

- Built a knowledge-extraction pipeline on the 200+ page PMBOK Guide, comparing 3 NLP normalization methods and 2 knowledge graph construction approaches (spaCy versus LLM) to produce a Neo4j graph with 597 nodes and 1,251 relationships.
- Developed a RAG chatbot leveraging Neo4j, HuggingFace embeddings, and Llama-3.2-90B via LangChain and ChatGrok, served through a Flask API and React frontend.
- Automated figure-to-text extraction using LlamaParse and LlamaIndex along with document parsing via PyMuPDF and OpenCV as part of a 6-person engineering team.

SKILLS AND LANGUAGES

Data Science & ML: Python, SQL, pandas, NumPy, scikit-learn, Random Forest, AdaBoost, Decision Trees, KMeans, DBSCAN, GMM, Hierarchical Clustering, GridSearchCV, Feature Engineering

AI, Deep Learning & LLMs: PyTorch, TensorFlow/Keras, HuggingFace Transformers, LangChain, NLP, Generative AI, RAG, Prompt Engineering

Web Development & Tools: Node.js, Flask, REST APIs, React, Angular, JavaScript, HTML/CSS, Git, Docker, CUDA, pytest

Databases & Analytics: SQL, MongoDB, MySQL, Oracle, Neo4j, Power BI, Streamlit, Matplotlib, Seaborn

Languages: Arabic (Native), English (C1), French (B2), German (A1), Spanish (A2)

EDUCATION

National Engineering Degree, Computer Science – Data Science Minor 09/2022 – 11/2025

ESPRIT · Tunis, Tunisia

- International program, fully taught in English.

Preparatory Institute for Engineering Studies of El Manar 09/2020 – 08/2022

Tunis, Tunisia

CERTIFICATES

Building RAG Agents with LLMs (NVIDIA) · **Building Transformer-Based NLP Applications (NVIDIA)** · **Introduction to Cloud Services (Microsoft Azure)**

INVOLVEMENT

Marketing Department Member September 2023 – November 2024

INCEPTUM Junior Enterprise

- Designed pitch decks, marketing reports, and analytical presentations to support partnership proposals and stakeholder communications.